

Emmanuel Hernandez

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EDUCATION

University of California, Los Angeles

Los Angeles, CA

Bachelor of Arts in Linguistics and Computer Science

Expected Graduation: June 2028

- **Relevant Coursework:** Data Structures & Algorithms, Algorithms & Complexity (in progress), Operating Systems, Computer Organization, Database Systems, Programming Languages, Software Construction & Tools, Computational Linguistics (in progress), Discrete Mathematics, Linear Algebra, Probability & Statistics, Multivariable Calculus

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, TypeScript, SQL, Swift, Rust

Frontend: React, React Native, Next.js, Expo, Tailwind CSS, HTML/CSS

Backend: Node.js, Express, REST APIs, Firebase, Firestore, PostgreSQL, Prisma, Supabase, MongoDB, JWT auth

Tools: Git, Linux/Unix, Docker, AWS, CI/CD, Vercel, Render, Postman, Google Apps Script, pytest

Agentic: Claude Code, MCP servers, custom subagents & skills, Ollama, LM Studio, Gemini API, LLM eval harnesses

EXPERIENCE

Software Engineering Intern

June 2026 – Aug 2026

Rise the Fenua

Los Angeles, CA

- Built a dashboard (Next.js, TypeScript, Supabase) used by 20+ volunteers to streamline merchandise fulfillment
- Engineered a data pipeline with 6+ SQL migrations to normalize Google Form responses, cutting manual processing by 40%
- Led operations across software, merchandise logistics, and fundraising that raised \$25,000 in donations
- Automated inventory and donation tracking with Google Apps Script, replacing manual shipment record-keeping

Software Engineer Intern

Dec 2025 – June 2026

We Explore Earth

Los Angeles, CA

- Developed features for a real-time event platform serving 20,000+ users (events, RSVPs, volunteer coordination)
- Designed 10+ REST APIs for event and RSVP workflows, consolidating sequential reads to cut redundant queries by 20%
- Built 20+ reusable React Native components from Figma hi-fis, working with 5 designers in a 15-member agile team
- Leveraged AWS services and Firebase-backed infrastructure to support scalable, real-time event management

PROJECTS

Database Crash Simulator | C, TypeScript, Bun, Rust, Linux

Aug 2026

- Found a confirmed data-loss bug in redb, a production Rust database, across 10,942 crash states tested in 21 minutes
- Built a crash-consistency fuzzer that replays legal post-crash disk states through a database's own recovery path
- Wrote an LD_PRELOAD C shim and TypeScript oracle pipeline, run on real ext4/xfs/btrfs images in a Linux VM
- Packaged a one-command reproducer running in 6 seconds, backed by 77 tests with SQLite as a control

LLM Code-Switching Benchmark | Python, pydantic, pytest, Ollama

Aug 2026 – Present

- Designed a 402-item benchmark testing if LLMs follow Matrix Language Frame theory in Spanish-English code-switching
- Evaluated 3 open-weight LLMs locally via Ollama; in 2, repair collapsed on bare English verbs in Spanish frames
- Engineered an eval harness (4 task scorers, bootstrap CIs, hash-keyed cache) with 177 tests at 96% coverage

SwipeBite | Next.js, React, Express, Prisma, PostgreSQL

Feb 2026 – Present

- Built a full-stack, Tinder-style restaurant discovery app with JWT auth, deployed live on Vercel and Render
- Wrote a hybrid recommender (content-based, collaborative filtering, Thompson Sampling) re-ranking the feed per swipe
- Engineered an Express 5 + Prisma/PostgreSQL API (7 route modules, 15 migrations) covered by 153 tests

Physics Remeshing Engine | C++

Jan 2025 – Apr 2025

- Built a C++ remeshing engine solving a second-order ODE by Euler's method over a spring-mass damping model
- Constrained interior points to rectangular, triangular and L-shaped regions using virtual boundary charges
- Selected to present at 2 undergraduate research conferences, including one hosted at UC Berkeley

LEADERSHIP & ORGANIZATIONS

Full-Stack Developer

Sep 2025 – Present

LA Blueprint

Los Angeles, CA

- Selected at a 4% acceptance rate for a student org building full-stack products for LA nonprofits
- Built and maintained React, Node.js/Express, and Firebase features shipped through reviewed pull requests
- Presented architecture proposals and ran sprint planning and design reviews across project teams
- Met with prospective nonprofit partners to scope web applications the club would build for them

ADDITIONAL

Awards: Dean's List; Southern California Research Symposium; Bay Area Honors Undergraduate Research Conference

Interests: Double Bass (music theory & performance), School on Wheels tutor